Pandora's Box & the Sins of Science

Before Eve and the apple, there was Pandora and her jar.

In Greek mythology, Zeus was furious with Prometheus for stealing fire and given it to humanity, so, he had the gods craft the first woman, Pandora. She was beautiful, clever, and irresistibly curious. She was sent to earth with a sealed jar and told never to open it:

As you can imagine, she opened it and out flew every affliction humanity would ever know. Everything from disease and suffering to envy and death. All of it released into the world with no way to call it back.

Only one thing remained trapped inside, as she slammed the lid closed: hope.

It's a story about good intentions, irresistible curiosity, and the impossibility of undoing what's already been done.

That myth is nearly three thousand years old. The story I'm going to relay today is that of something much more recent.

A pediatrician named Dr. Paul Offit, a vaccine researcher who spent his career studying how good science goes bad, discovered the Pandora theme continuing to recur throughout the history of science.

In his 2017 book titled Pandora's Lab, Offit tells seven true stories of scientific breakthroughs that on the surface appear to be those of unmistakable progress, until each one revealed a darker truth once it was released upon the world.

Two of those stories almost perfectly illustrate that Pandora myth.

The first is the invention of the lobotomy in the mid 1930's, and the other is that of synthetic fertilizer.

Two very different breakthroughs, two very different scientists, same story playing out: curiosity opening the jar of mystery with the best of intentions, and the evils that were unleashed and could never be called back.

Now, most people have heard of Jeffrey Dahmer.

What you may not know is that decades before he became one of the most infamous names in America, one of the very techniques associated with his crimes had, 50 years earlier, earned someone else a Nobel Prize.

What began in a hospital as a celebrated medical breakthrough would, within a generation, become inseparable from horror.

According to Offit's book, in 1935, Egas Moniz (**EH-guzh - moo-NEESH**), a Portuguese neurologist, attended a lecture where two Yale scientists described an experiment on chimpanzees. By removing their frontal lobes, they were able to turn anxious, aggressive animals into docile ones.

Moniz needed no further convincing.

Within months, he found a surgeon to help him open the jar. They drilled into a human brain for the first time, with no animal trials and no long-term data, only the desperate hope of relieving suffering. He called it a great step forward, and fourteen years later, he won the Nobel Prize for it.

A year later an American physician named Walter Freeman took Moniz's idea and turned it into a nationwide phenomenon — eventually performing lobotomies in his office with an ice pick borrowed from his own kitchen. Tens of thousands of patients emerged not cured, but diminished and childlike, all in the name of relief.

Like Pandora, Moniz and Freeman weren't villains chasing harm. They were chasing hope. And like Pandora, once that jar was opened, the demons couldn't be put back.

Now, everyone knows about the gas chambers at Auschwitz, but what you may not know is that decades earlier, the same man responsible for the chemistry that created that horror, had already saved billions of lives from starvation.

As Offit again explains, in 1909, Fritz Haber solved a problem that had been stumping scientists for a century: how to pull nitrogen from the air and turn it into fertilizer.

Before his discovery, scientists were predicting mass starvation by the 1930s. The soil just didn't have enough natural fertilizer to feed the growing population.

But that same chemistry also produced explosives, and Haber's factories ultimately armed Germany's war machine through World War I. Haber even went on to personally oversee the first large-scale poisonous gas attacks in history and pioneered chemical warfare.

Decades later, his lab’s pesticide technology was re-engineered into Zyklon B, which was used by the Nazis to murder over a million people — including members of Haber's own family.

Again, one man finding hope and death sealed in the same jar.

But Pandora's myth isn’t meant to end in despair — it’s meant to end with hope, still there after everything else has spilled out. And that's true of both these stories as well.

Walter Freeman's icepick lobotomies were a monstrous detour, but the backlash against them helped push psychiatry to keep searching for real, working treatments for mental illness.

Fritz Haber's chemistry gave the world chemical warfare and, indirectly, the gas chambers, but it also feeds nearly half the people on the planet right now.

The lesson here isn't that science is dangerous, or that curiosity is something to fear. It's that every real breakthrough is a jar we can't fully see inside of before we open it. The evils and the hopes come out together, and we don't get to choose only one. What we can choose is to stay skeptical of the quick solutions and ask what else might be inside before we celebrate.